

Key Concept of CESD website captured

1. Vision / Mission / Objectives

CESD will be a national hub for engineering innovation, academic–industry collaboration, sustainable industrial growth, research, testing, training, and technical services under FEN-NUOL.

2. Main services

CESD services should include:

- Student project support
- Curriculum-based laboratory practice
- Faculty research support
- Contracted analysis/testing for companies
- Joint R&D with industry
- Talent development and training
- Facility/equipment sharing
- Customized training and workshops
- Technical seminar / meeting room / exhibition support

3. Laboratories and facilities

CESD has the following core facilities:

- Chemical Analysis Laboratory
- Electron Microscope Laboratory
- X-ray Analysis Laboratory
- High-Voltage Laboratory
- Solid Forming Laboratory
- Computer Room
- Physical Measurement Laboratory
- Electrical and Electronic Measurement Laboratory
- Meeting Room
- Exhibition Room
- Administration Office

The Lao document also mentions more than **300 laboratory equipment items** supported by JICA.

5. Staff / management structure

The proposed CESD organization has:

- Center Director
- Deputy Director
- Administration and Planning Unit
- Technical Service and Laboratory Unit
- Innovation, AI Database and Training Unit
- Laboratory technicians / technical officers
- Finance and administration staff

Recommended Laravel/Next.js System Direction

For CESD, I recommend this system structure:

Frontend: Next.js
Backend/API: Laravel
Database: PostgreSQL or MySQL
Admin Panel: Laravel Filament
Authentication: Laravel Sanctum
File Storage: Local server or S3-compatible storage
Languages: Lao / English

Main Website Modules

1. Public Website
 - Home
 - About CESD
 - Vision & Mission
 - Laboratories
 - Equipment Catalog
 - Services
 - Training
 - Projects / Partners
 - News & Events
 - Contact
2. Service Platform
 - Request testing service
 - Request consulting service
 - Request training
 - Request room/equipment use
 - Track request status
3. Equipment & Laboratory Management
 - Equipment catalog
 - Equipment category
 - Laboratory location
 - Responsible staff
 - Booking schedule
 - Maintenance / calibration record
4. Admin Dashboard
 - Manage content
 - Manage services
 - Manage requests
 - Manage equipment
 - Manage users
 - Manage reports
 - View statistics

Proposed First Development Phase

Build Phase 1 MVP first:

- Bilingual public website

- Laboratory directory
- Equipment catalog
- Service request form
- Training request form
- News/events management
- Admin dashboard

Then **Phase 2:**

- Online booking calendar
- Quotation management
- Customer CRM
- QR-code report verification
- Digital report upload/download
- Maintenance and calibration tracking
- Analytics dashboard

Next, please provide item **4: departments/expert teams involved**, and any available **equipment list** if you have it. Then I can prepare the full **Database Design + Use Case Diagram + Developer Prompt for Laravel/Next.js**.